



The Scots College

HSC Mathematics Extension 2

Pre Trial Examination

April 2013

General Instructions

- Working time : 2 hours + 5 minutes reading time.
- Write using blue or black pen
- Board approved calculators may be used (Non Graphic)
- All necessary working should be shown in every question
- Standard Integrals Table attached
- Answer each question on a SEPARATE answer booklet

TOTAL MARKS: 75

SECTION I

7 marks

- Attempt Questions 1-7
- Answer on the Multiple Choice answer sheet provided.
- Allow about 10 minutes for this section.

SECTION II

68 marks

- Attempt questions 8 – 11
- Answer on the booklets provided, unless otherwise instructed. Start a new booklet for each question.
- Allow about 1 hour & 50 minutes for this section

WEIGHTING: 30 %

Section I**7 marks****Attempt Questions 1-7****Allow about 10 minutes for this section**

Use the multiple choice answer sheet for questions 1 – 7.

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample $2 + 4 = ?$ (A) 2 (B) 6 (C) 8 (D) 9A ☐ B ☒ C ☐ D ☐

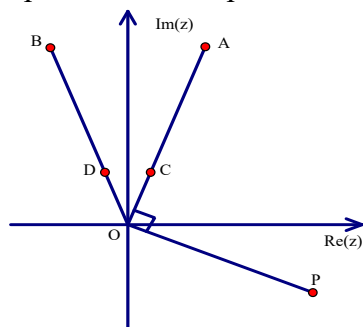
If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ☒ B ☒ C ☐ D ☐If you change your mind and have crossed out what you consider to be the correct answer, then indicate this by writing the word *correct* and drawing an arrow as follows:A ☒ B ☒ C ☐ D ☐
correct

1 Let $z^2 = \sqrt{3}i$, what is the value of z^{8n+4} ?

- A) -3^{2n+1} B) $(-3)^{8n+2}$ C) $(-3)^{6n+4}$ D) 3^{2n+1}

2 The point P represents the complex number ω where $\omega\bar{\omega} = 4$. Which of the points A, B, C, or D shown in the Argand diagram could represent the complex number $i\omega^{-1}$?



3 For the curve with equation $y = [f(x)]^n$:

- (A) The x intercept(s) of $y = f(x)$ correspond to the stationary points of the curve.
- (B) The curve exhibits point symmetry about the origin.
- (C) The curve does not exist for values for which $f(x) < 0$.
- (D) More information is needed about the curve to determine further properties.

- 4 The roots of the equation $x^3 - 3x + 3 = 0$ are α, β and γ . What is the value of $\alpha^2 + \beta^2 + \gamma^2$.

A) -6 B) -3 C) 3 D) 6

- 5 The Cartesian equation of the locus of complex number z , such that $|z + 3| + |z - 3| = 12$, is

(A) $\frac{x^2}{36} - \frac{y^2}{27} = 1$

(B) $\frac{x^2}{36} + \frac{y^2}{27} = 1$

(C) $\frac{x^2}{27} - \frac{y^2}{36} = 1$

(D) $\frac{x^2}{27} + \frac{y^2}{36} = 1$

- 6 The foci of the ellipse $25x^2 + 9y^2 = 225$ have the coordinates

A) $(\pm 4, 0)$ B) $(\pm\sqrt{34}, 0)$ C) $(0, \pm 4)$ D) $(0, \pm\sqrt{34})$

- 7 The eccentricity of the hyperbola $x^2 - y^2 = 4$ is

A) 2 B) $\frac{1}{2}$ C) $\frac{1}{\sqrt{2}}$ D) $\sqrt{2}$

End of Section I

Section II**Total Marks (68)****Attempt Questions 8 – 11.****Allow about 1 hours & 50 minutes for this section.**

Answer all questions, starting each question on a new answer booklet with your name and question number at the top of the page.

All necessary working should be shown in every question.

Question 8 (Marks 17) Answer on a new booklet.

a) Evaluate i^{2054} [1]

b) Let $z = \frac{9 + 2i}{2 + i}$. [4]

i) Simplify $(9 + 2i)(\overline{2 + i})$

ii) Express z in the form $a + ib$, where a and b are real numbers.

iii) Hence, or otherwise, find $|z|$ and $\arg(z)$.

c) If $z = \cos \theta + i \sin \theta$ [2]
Show that $1 + z = 2 \cos \frac{\theta}{2} \left(\cos \frac{\theta}{2} + i \sin \frac{\theta}{2} \right)$.

d) Let $z = \cos \theta + i \sin \theta$. [4]
Given that $z^n + \frac{1}{z^n} = 2 \cos n\theta$
and that $z^n - \frac{1}{z^n} = 2i \sin n\theta$
Show that $\sin^4 \theta + \cos^4 \theta = \frac{1}{4} (\cos 4\theta + 3)$

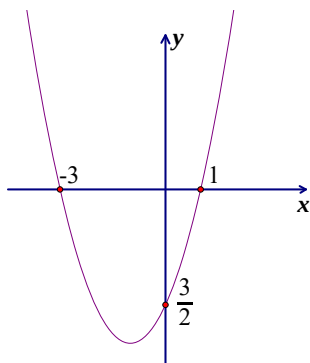
e) Express $(3 + 2i)(5 + 4i)$ and $(3 - 2i)(5 - 4i)$ in the form $x + iy$. [3]
Hence express $7^2 + 22^2$ as a product of prime factors.

f) On an Argand diagram, shade in the region containing all points [3]
representing the complex number z , such that
 $|z - (2 + 2i)| \leq 2$ and $|z - (2 + 2i)| \leq |z|$

Question 9 (Marks 17) Answer on a new booklet.

[7]

a)



The sketch above shows the parabola

$$y = f(x) \quad \text{where} \quad f(x) = \frac{1}{2}(x^2 + 2x - 3)$$

Without the use of calculus, draw separate sketches of the following, showing all important features such as intercepts, asymptotes and turning points.

- i) $y^2 = f(x)$
- ii) $y = e^{f(x)}$
- iii) $y = \tan^{-1}f(x)$
- iv) $y = \frac{1}{2}(x+3)|x-1|$

b) Consider the graph of the function $y = \frac{x^2}{x+2}$ [7]

- i) Find the equations of the asymptotes for the graph.
- ii) Find the stationary points and state their nature
- iii) Draw a neat sketch of the graph, showing all important features.
- iv) Hence, or otherwise, sketch the graph of $y = \frac{|x+2|}{x^2}$

c) Find the coordinates of the stationary points and determine their nature, for the curve [3]

$$x^2 + xy - 2y^2 + 9 = 0$$

Question 10 (Marks 17) Answer on a new booklet.

a) Consider the hyperbola $\frac{x^2}{16} - \frac{y^2}{9} = 1$. Find [7]

- i. its eccentricity
- ii. the coordinates of the foci
- iii. equation of the directrices
- iv. equation of the asymptotes
- v. the length of the latus rectum
- vi. draw a neat sketch of the hyperbola, showing all the important features.

b) The point $P \left(4p, \frac{4}{p} \right)$, lies on the hyperbola $xy = 16$ and is in the first quadrant. The normal to the hyperbola at P meets the hyperbola again at the point $Q \left(4q, \frac{4}{q} \right)$. [6]

- i. Find the equation of the normal at P .
- ii. Show that $q = -\frac{1}{p^3}$
- iii. Hence show that there is only one value of p for which the normal at P is also a normal to the hyperbola at Q . Find the coordinates of P and Q for this normal.

c) Let $C_1 = x^2 + 4y^2 - 2$ and $C_2 = 3x^2 + y^2 - 1$, and let k be a real number. [4]

- i. Show that $C_1 + k C_2 = 0$ is the equation of the curve passing through the point of intersection of the ellipses $C_1 = 0$ and $C_2 = 0$.
- ii. Find the value of k such that $C_1 + k C_2 = 0$ is the equation of an ellipse.

Question 11 (Marks 17) Answer on a new booklet.

- a) Prove by Mathematical Induction that [4]

$$\frac{(3+\sqrt{5})^n + (3-\sqrt{5})^n}{2^n} \text{ is a positive integer, for all positive integer values of } n.$$

You may assume that the statement is true for $n = k$ and $n = k + 1$, and that

$$a^{n+1} + b^{n+1} = (a + b)(a^n + b^n) - ab(a^{n-1} + b^{n-1})$$

- b) Show that $x^5 - 5x + 1 = 0$ cannot have a double root. [3]

- c) Solve the equation [4]

$$8x^4 - 14x^3 - 69x^2 - 14x + 8 = 0$$

- d) (i) Solve the equation [6]

$$x^6 - x^3 + 1 = 0$$

(ii) Hence express $x^6 - x^3 + 1$ as a product of three real quadratic factors.

(iii) Hence show that

$$\cos \frac{\pi}{9} + \cos \frac{7\pi}{9} + \cos \frac{13\pi}{9} = 0, \text{ and}$$

$$\cos \frac{\pi}{9} \times \cos \frac{7\pi}{9} \times \cos \frac{13\pi}{9} = \frac{1}{8}$$

End of Assessment

Standard Integrals

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

Pre-Trial HSC Examination 2013
Mathematics Extension 2

Multiple Choice Answer Sheet

Name _____

Completely fill the response oval representing the most correct answer.

- | | | | | | | | | |
|----|---|-----------------------|---|-----------------------|---|-----------------------|---|-----------------------|
| 1. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 2. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 3. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 4. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 5. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 6. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 7. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |

Pre-Trial HSC Examination 2013
Mathematics Extension 2

Multiple Choice Answer SheetName SOLUTIONS.

Completely fill the response oval representing the most correct answer.

- | | | | | | | | | |
|----|---|----------------------------------|---|----------------------------------|---|----------------------------------|---|----------------------------------|
| 1. | A | ☒ | B | ☐ | C | ☐ | D | ☐ |
| 2. | A | ☐ | B | ☐ | C | ☐ | D | ☒ |
| 3. | A | ☒ | B | ☐ | C | ☐ | D | ☒ |
| 4. | A | ☐ | B | ☐ | C | ☐ | D | ☒ |
| 5. | A | ☐ | B | ☒ | C | ☐ | D | ☐ |
| 6. | A | ☐ | B | ☐ | C | ☒ | D | ☐ |
| 7. | A | ☐ | B | ☐ | C | ☐ | D | ☒ |

Both answers acceptable

Question _____

Name: _____

Teacher: _____

YEAR 12 MATHEMATICS EXTENSION 2

PRE-TRIALS SOLUTIONS - 2013

Q1. $z^2 = \sqrt{3}i$

$$z^{8n+4} = (z^2)^{4n+2}$$

$$= (\sqrt{3}i)^{4n+2}$$

$$= [(\sqrt{3}i)^2]^{2n+1}$$

$$= (-3)^{2n+1}$$

(A)

2. $i\omega^{-1} = \frac{i}{\omega} = \frac{i\omega}{\omega\omega} = \frac{i\omega}{4}$

(D)

3. (A)

$$\begin{aligned} 4. \alpha^2 + \beta^2 + \gamma^2 &= (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \beta\gamma + \gamma\alpha) \\ &= 0 - (-3) \\ &= 3 \end{aligned}$$

(D)

Question _____

Name: _____

Teacher: _____

Q5) $|z-3| + |z+3| = 12$

$$a = 6 \quad a^2 = 36$$

$$ae = 3, \quad e = \frac{1}{2}$$

$$b^2 = 36(1 - \frac{1}{4})$$

$$= 27$$

(B)

Q6 $25x^2 + 9y^2 = 225$

$$\frac{x^2}{9} + \frac{y^2}{25} = 1$$

$$a^2 = 25 \quad b^2 = 9$$

$$9 = 25(1 - e^2)$$

$$1 - e^2 = \frac{9}{25}$$

$$e^2 = 1 - \frac{9}{25}$$

$$= \frac{16}{25}$$

$$e = \frac{4}{5} \quad ae = 4$$

(C)

Q7

(D)

Question _____

Name: _____

Teacher: _____

SECTION II

Q 8.

(a)

$$\begin{aligned} & \cdot 2054 \\ & \cdot 4(513) + 2 \\ & = \cdot 2 \\ & = \cdot 2 \\ & = \textcircled{-1} \end{aligned}$$

(b) (i) $(9+2i)(\overline{2+i})$

$$= (9+2i)(2-i)$$

$$= 18 + 2 - 9i + 4i$$

$$= 20 - 5i$$

(ii) $z = \frac{9+2i}{2+i}$

$$= \frac{(9+2i)(2-i)}{(2+i)(2-i)}$$

$$= \frac{20-5i}{5} = 4-i$$

(iii) $|z| = \sqrt{16+1} = \sqrt{17}$

$$\arg(z) = \tan^{-1}\left(-\frac{1}{4}\right)$$

Question _____

Name: _____

Teacher: _____

$$(c) \quad Z = \cos \theta + i \sin \theta$$

$$1+Z = 1 + \cos \theta + i \sin \theta$$

$$= 2 \cos^2 \frac{\theta}{2} + 2i \sin \frac{\theta}{2} \cos \frac{\theta}{2}$$

$$= 2 \cos \frac{\theta}{2} (\cos \frac{\theta}{2} + i \sin \frac{\theta}{2})$$

$$(d) \quad Z^n + \frac{1}{Z^n} = 2 \cos n\theta$$

$$Z^n - \frac{1}{Z^n} = 2i \sin n\theta$$

let $n=1$

$$(2i \sin \theta)^4 = \left(Z - \frac{1}{Z}\right)^4$$

$$16 \sin^4 \theta = Z^4 - 4Z^2 + 6 - \frac{4}{Z^2} + \frac{1}{Z^4} \quad \text{--- (1)}$$

$$(2 \cos \theta)^4 = \left(Z + \frac{1}{Z}\right)^4$$

$$16 \cos^4 \theta = Z^4 + 4Z^2 + 6 + \frac{4}{Z^2} + \frac{1}{Z^4}$$

$$16 (\sin^4 \theta + \cos^4 \theta) = 2Z^4 + 12 + \frac{2}{Z^4}$$

$$= 2(2 \cos 4\theta) + 12$$

$$= 4(\cos 4\theta + 3)$$

$$\therefore \sin^4 \theta + \cos^4 \theta = \frac{1}{4} (\cos 4\theta + 3)$$

Question _____

Name: _____

Teacher: _____

$$(e) (3+2i)(5+4i) = 15 - 8 + 12i + 10i \\ = 7 + 22i$$

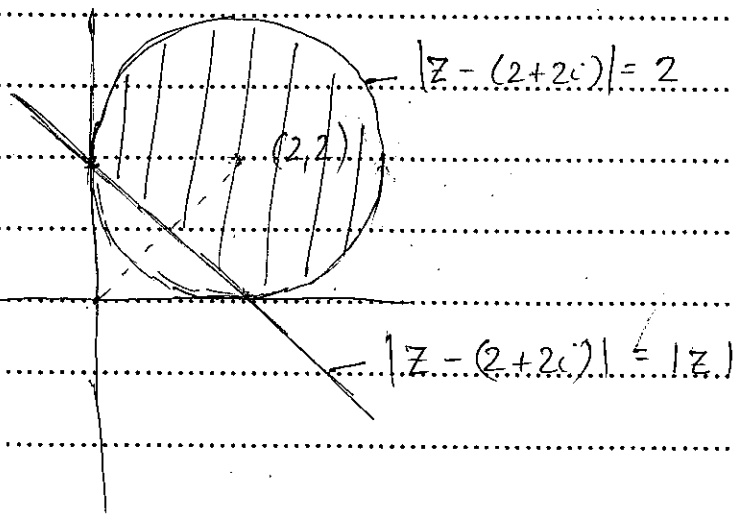
$$(3-2i)(5-4i) = 15 - 8 - 12i - 10i \\ = 7 - 22i$$

$$(3+2i)(5+4i)(3-2i)(5-4i) = (7+22i)(7-22i)$$

$$(9+4)(25+16) = 7^2 + 22^2$$

$$\text{or } 7^2 + 22^2 = 13 \times 41$$

(f)

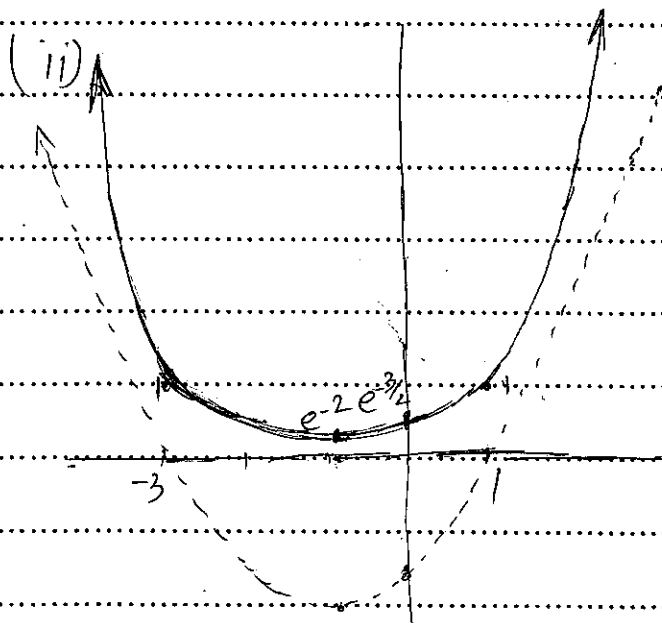
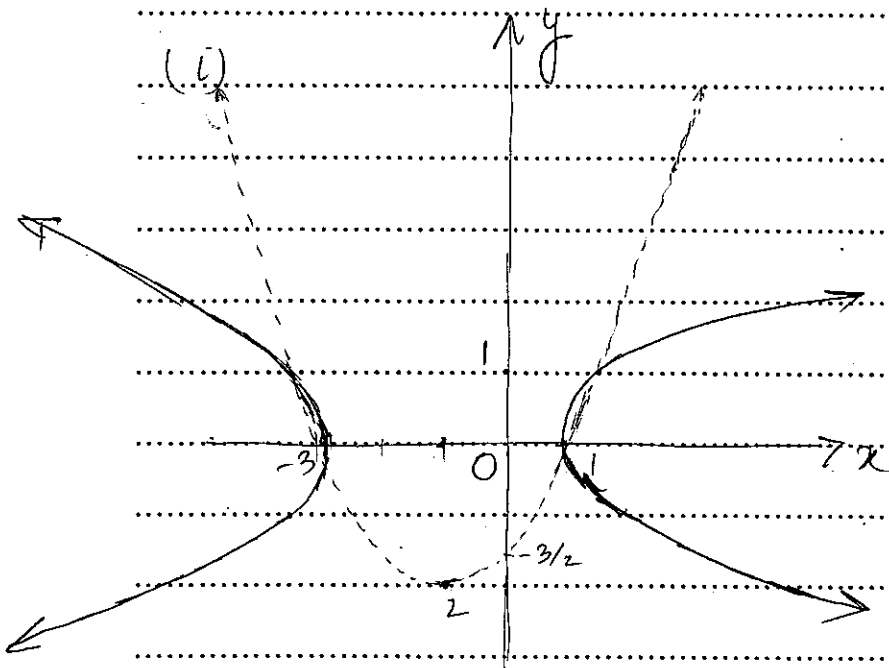


Question _____

Name: _____

Teacher: _____

Q. 9.

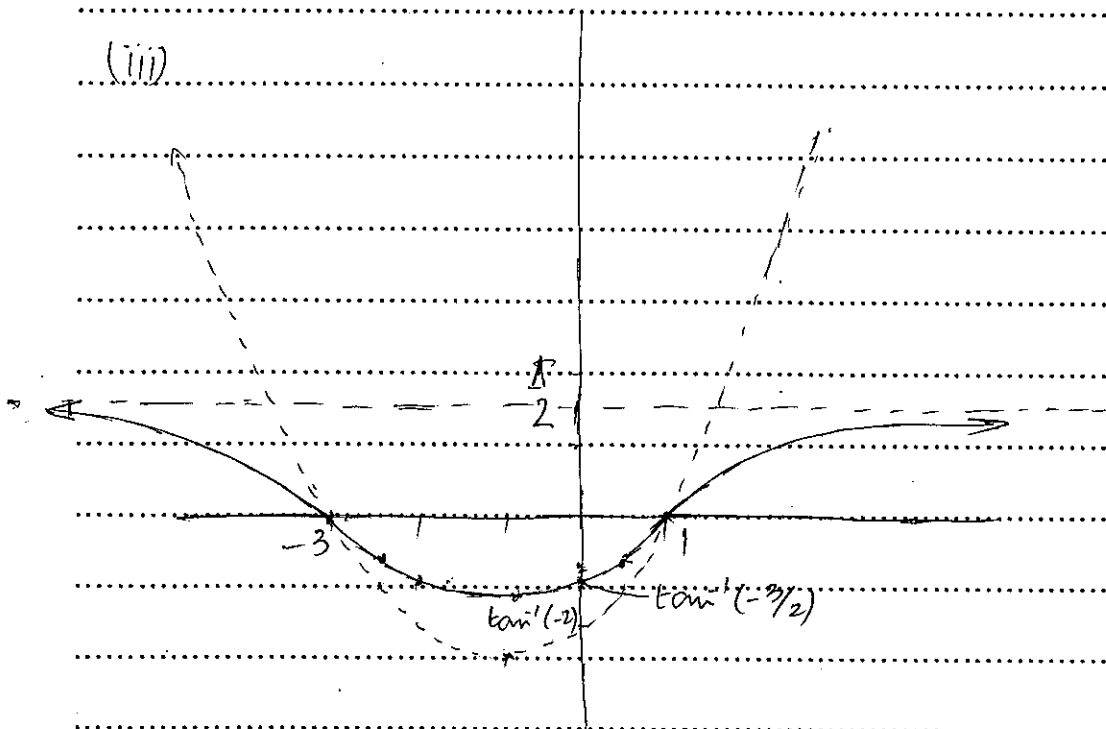


Question _____

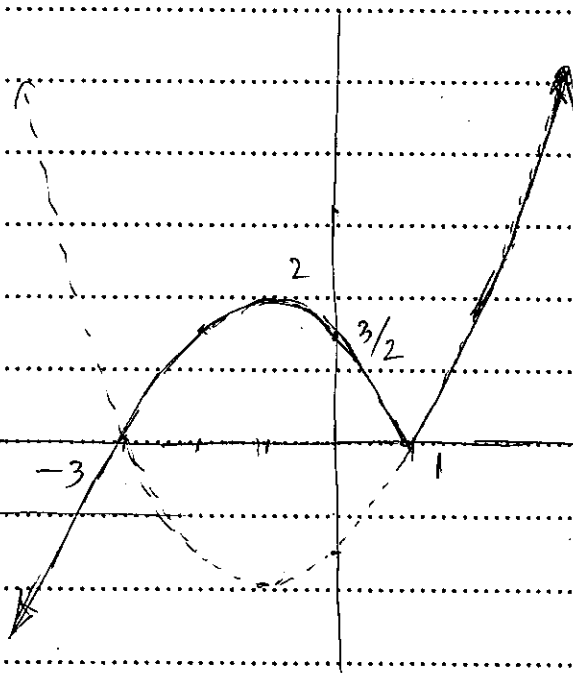
Name: _____

Teacher: _____

(iii)



(iv)



$$y = \frac{1}{2}(x+3)(x-1)$$

$$= \frac{1}{2}(x+3)(x-1)$$

for $x \geq 1$

$$y = 2 - \frac{1}{2}(x+3)(x-1)$$

for $x < 1$

Question _____

Name: _____

Teacher: _____

9(b)

$$y = \frac{x^2}{x+2}$$

$$(i) \quad y = \frac{x^2 - 4 + 4}{x+2}$$

$$= x - 2 + \frac{4}{x+2}$$

Oblique asymptote at $y = x - 2$

Vertical asymptote $x = -2$

$$(ii) \quad \frac{dy}{dx} = \frac{(x+2)(2x) - x^2(1)}{(x+2)^2}$$

$$= \frac{2x^2 + 4x - x^2}{(x+2)^2}$$

$$= \frac{x^2 + 4x}{(x+2)^2}$$

$$\frac{d^2y}{dx^2} = 1 - \frac{4}{(x+2)^2}$$

$$\frac{d^2y}{dx^2} = \frac{8}{(x+2)^3}$$

for stationary points $\frac{dy}{dx} = 0$

$$1 = \frac{4}{(x+2)^2} \quad (x+2)^2 = 4$$

$$x+2 = \pm 2$$

$$x = 0 \text{ or } -4$$

$$y = 0 \text{ or } -8$$

When $x = 0$, $\frac{d^2y}{dx^2} > 0$

$\therefore (0, 0)$ is min. pt

$x = -4$, $\frac{d^2y}{dx^2} < 0$

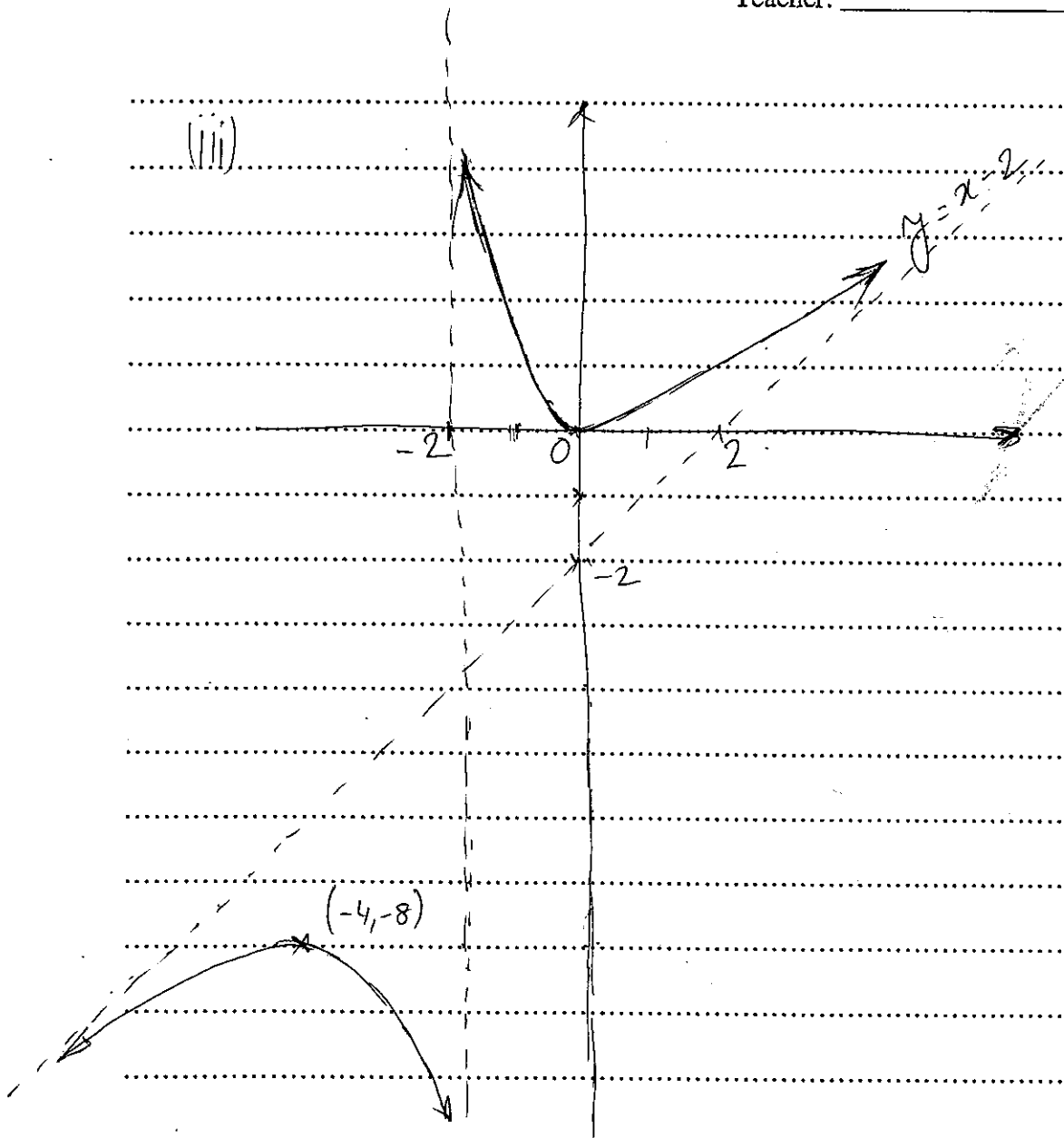
$\therefore (-4, -8)$ is max. pt

Question _____

Name: _____

Teacher: _____

(iii)



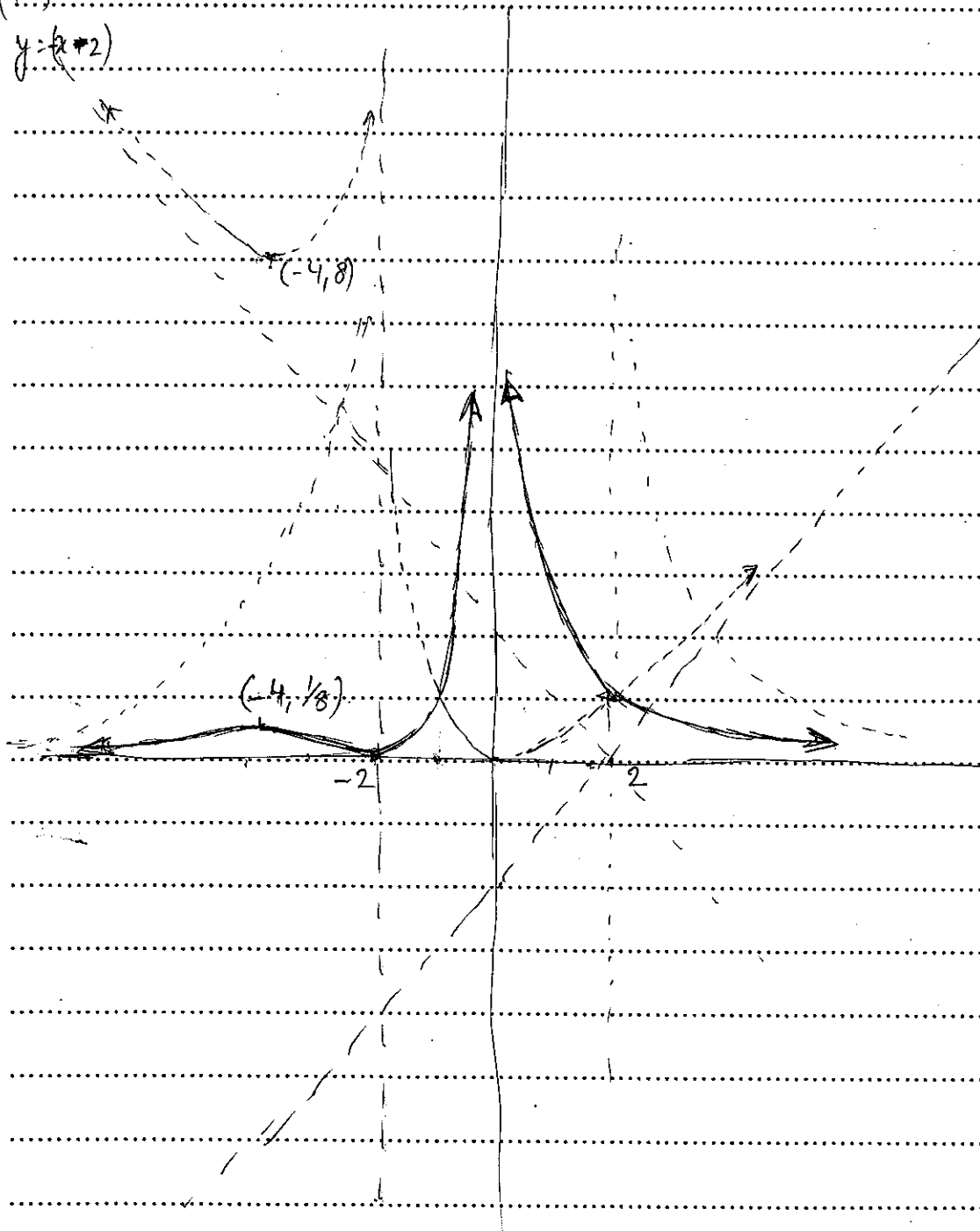
Question _____

Name: _____

Teacher: _____

(IV)

$y = (x+2)$



Question _____

Name: _____

Teacher: _____

9(c)

$$x^2 + xy - 2y^2 + 9 = 0$$

Differentiating w.r.t. x

$$2x + y + x y' - 4y y' = 0$$

$$2x + y = (4y - x) y'$$

$$y' = \frac{2x + y}{4y - x}$$

$$y' = \frac{(4y - x)(2 + y') - (2x + y)(4y' - 1)}{(4y - x)^2}$$

For st. points $y' = 0$ When $x = 1, y = -2, y' = 0$

$$2x + y = 0$$

$$y'' = \frac{(-8 - 1)(2 + 0) - 0}{(-8 - 1)^2}$$

$$y = -2x$$

 $< 0 \therefore (1, -2)$ is a max ptSub into $x^2 + xy - 2y^2 + 9 = 0$

$$x^2 - 2x^2 - 2(4x^2) + 9 = 0$$

When $x = -1, y = 2, y' = 0$

$$-9x^2 + 9 = 0$$

$$y'' = \frac{(8 + 1)(2 + 0) - 0}{(8 + 1)^2}$$

$$x^2 = 1$$

 > 0

$$x = 1, \text{ or } x = -1$$

$$y = -2 \text{ or } y = 2$$

 $\therefore (-1, 2)$ is a min. pt

Question _____

Name: _____

Teacher: _____

10

$$(a) \quad \frac{x^2}{16} - \frac{y^2}{9} = 1$$

$$a^2 = 16, \quad b^2 = 9$$

$$9 = 16(e^2 - 1)$$

$$e^2 - 1 = \frac{9}{16}$$

$$e^2 = 1 + \frac{9}{16} = \frac{25}{16}$$

$$(i) \quad e = \frac{5}{4}$$

$$(ii) \quad \text{foci: } ae = 4 \times \frac{5}{4} = 5$$

$$\therefore \text{foci: } (\pm 5, 0)$$

$$(iii) \quad \text{Equation of directrices: } x = \pm \frac{a}{e}$$

$$x = \pm \frac{16}{5}$$

(iv) Asymptotes

$$y = \pm \frac{bx}{a}$$

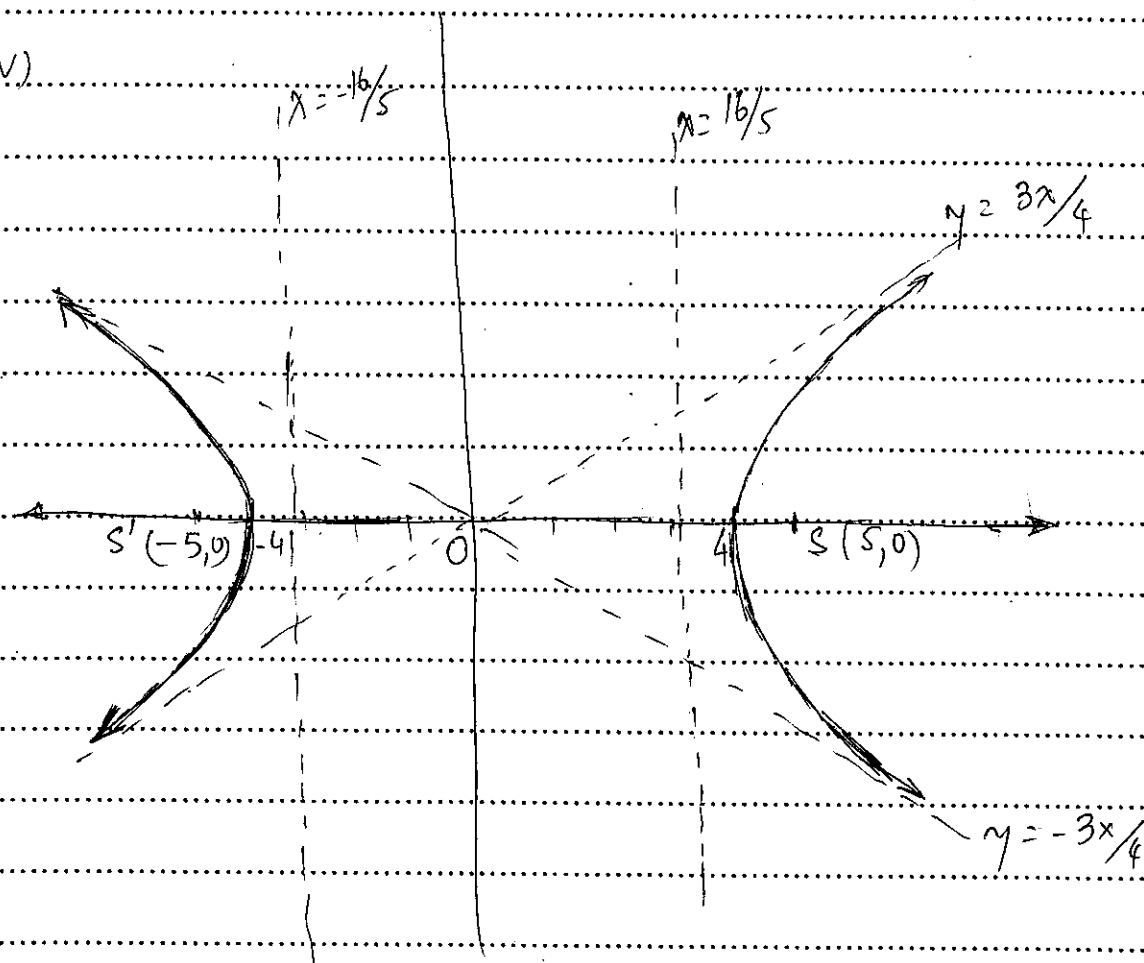
$$y = \pm \frac{3x}{4}$$

Question _____

Name: _____

Teacher: _____

(V)



Question _____

Name: _____

Teacher: _____

10(b)

$$xy = 16$$

$$(i) \quad y = 16/x$$

$$\frac{dy}{dx} = -\frac{16}{x^2}$$

$$\text{at } p, m_t = -\frac{16}{16p^2} = -\frac{1}{p^2}$$

$$m_N = p^2$$

Equation of normal is

$$y - \frac{4}{p} = p^2(x - 4p)$$

$$\text{or } py - 4 = p^3(x - 4p)$$

$$\text{or } p^3x - py = 4p^4 - 4$$

(ii) Sub $(4q, 4/q)$ into the equation of the normal

$$p^3(4q) - \frac{4p}{q} = 4p^4 - 4$$

$$4p^3q^2 - 4p = 4p^4q - 4q$$

$$p^3q^2 - p^4q = p - q$$

$$p^3q(q - p) = p - q \Rightarrow p^3q = -1 \therefore q = -\frac{1}{p^3}$$

Question _____

Name: _____

Teacher: _____

$$(iii) M_N \text{ at } Q = q^2 = M_N \text{ at } P.$$

$$\therefore p^2 = q^2$$

$$= \left(\frac{-1}{p^3} \right)^2$$

$$\text{or } p^2 = \frac{1}{p^6}$$

$$p^8 = 1$$

$$p^4 = 1 \text{ or } -1 (\text{rej})$$

$$p^2 = 1 \text{ or } -1 (\text{rej})$$

$$\underline{\underline{p = 1}} \text{ or } -1 (\text{rej as } P \text{ is in } 1^{\text{st}} \text{ quadrant})$$

$$\therefore P(4, 4) \quad , \quad Q(-4, -4)$$

Question _____

Name: _____

Teacher: _____

$$(C) \quad C_1 \equiv x^2 + 4y^2 - 2 \quad C_2 \equiv 3x^2 + y^2 - 1$$

(i) The pt. of intersection of C_1 and C_2 must satisfy $C_1 = 0$ and $C_2 = 0$

It will satisfy

$$C_1 + k C_2 = 0$$

$\therefore C_1 + k C_2 = 0$ is the required equation

$$(ii) \quad x^2 + 4y^2 - 2 + k(3x^2 + y^2 - 1) = 0$$

$$\text{or } (1+3k)x^2 + (4+k)y^2 - 2-k = 0$$

$$\text{or } \frac{x^2}{\frac{(2+k)}{1+3k}}} + \frac{y^2}{\frac{(2+k)}{4+k}}} = 1$$

For an ellipse

$$\frac{2+k}{1+3k} > 0, \quad \frac{2+k}{4+k} > 0 \quad \text{and} \quad \frac{2+k}{1+3k} \neq \frac{2+k}{4+k}$$

$$(2+k)(1+3k) > 0$$

$$k < -2 \text{ or } k > -\frac{1}{3}$$

$$(2+k)(4+k) > 0$$

$$k > -2 \text{ or } k < -4$$

$$1+3k \neq 4+k$$

$$2k \neq 3$$

$$k \neq \frac{3}{2}$$

$$\therefore \underline{k > -\frac{1}{3}, \quad k < -4, \quad k \neq \frac{3}{2}}$$

Question _____

Name: _____

Teacher: _____

Q 11

(a) $\frac{(3+\sqrt{5})^n + (3-\sqrt{5})^n}{2^n}$ is a positive integer

Step 1 prove true for $n=1$ and $n=2$

$$\frac{(3+\sqrt{5})^1 + (3-\sqrt{5})^1}{2^1} = \frac{6}{2} = 3$$

$$\frac{(3+\sqrt{5})^2 + (3-\sqrt{5})^2}{2^2} = \frac{9+5+6\sqrt{5}+9+5-6\sqrt{5}}{4}$$

$$= \frac{28}{4}$$

$$= 7$$

Hence true for $n=1$ and $n=2$

Step 2 Assume true for $n=k$ and $n=k+1$

$$\frac{(3+\sqrt{5})^k + (3-\sqrt{5})^k}{2^k} = p \quad \text{and} \quad \frac{(3+\sqrt{5})^{k+1} + (3-\sqrt{5})^{k+1}}{2^{k+1}} = q$$

where p and q are positive integers

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Step 3 Prove true for $n = k+2$

ie $\frac{(3+\sqrt{5})^{k+2} + (3-\sqrt{5})^{k+2}}{2^{k+2}}$ is an integer

$$\frac{(3+\sqrt{5})^{k+2} + (3-\sqrt{5})^{k+2}}{2^{k+2}} = \frac{(3+\sqrt{5} + 3-\sqrt{5})((3+\sqrt{5})^{k+1} + (3-\sqrt{5})^{k+1}) - (3+\sqrt{5})(3-\sqrt{5})(3+\sqrt{5})^k + (3-\sqrt{5})(3+\sqrt{5})^k}{2^{k+2}}$$

$$= \frac{6(q \cdot 2^{k+1}) - (q+\sqrt{5})(p \cdot 2^k) - (q-\sqrt{5})(p \cdot 2^k)}{2^{k+2}}$$

$$= \frac{3q \cdot 2^{k+2} - p \cdot 2^{k+2}}{2^{k+2}}$$

$$= 3q - p$$

which is an integer since p & q are positive integers.

Hence true for $n = k+2$

Step 4: Therefore by the principle of mathematical induction, the statement is true for all positive integers n .

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$$(b) \quad x^5 - 5x + 1 = 0$$

$$\text{let } P(x) = x^5 - 5x + 1$$

$$P'(x) = 5x^4 - 5 = 0$$

$$x^4 = 1$$

$$x = 1, -1, i, -i$$

$$P(1) = 1 - 5 + 1 \neq 0$$

$$P(-1) = -1 + 5 + 1 \neq 0$$

$$P(i) = i - 5i + 1 \neq 0$$

$$P(-i) = -i + 5i + 1 \neq 0$$

$\therefore$ No root of $P'(x) = 0$ is a root of $P(x) = 0$

$\therefore$ No double root

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$$(c) \quad 8x^4 - 14x^3 - 69x^2 - 14x + 8 = 0$$

$$\div x^2$$

$$8x^2 - 14x - 69 - \frac{14}{x} + \frac{8}{x^2} = 0$$

$$\text{or } 8\left(x^2 + \frac{1}{x^2}\right) - 14\left(x + \frac{1}{x}\right) - 69 = 0$$

$$\text{let } t = x^2 + \frac{1}{x^2}$$

$$t^2 = x^2 + 2 + \frac{1}{x^2} \Rightarrow t - 2 = x^2 + \frac{1}{x^2}$$

$$\therefore 8(t-2) - 14(t) - 69 = 0$$

$$\text{or } 8t^2 - 14t - 85 = 0$$

$$8 \times 85$$

$$8t^2 - 34t + 20t - 85 = 0$$

$$= 4 \times 2 \times 17 \times 5$$

$$2t(4t-17) + 5(4t-17) = 0$$

$$= 34, 20$$

$$\text{or } (2t+5)(4t-17) = 0$$

$$\therefore t = -5/2 \text{ or } 17/4$$

$$x + \frac{1}{x} = -\frac{5}{2}$$

$$x + \frac{1}{x} = \frac{17}{4}$$

$$2x^2 + 2 = -5x$$

$$4x^2 + 4 = 17x$$

$$\text{or } 2x^2 + 5x + 2 = 0$$

$$4x^2 - 17x + 4 = 0$$

$$(2x+1)(x+2) = 0$$

$$(4x-1)(x-4) = 0$$

$$x = -\frac{1}{2} \text{ or } -2$$

$$x = \frac{1}{4} \text{ or } 4$$

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$$(d) \quad x^6 - x^3 + 1 = 0$$

$$x^3 = \frac{1 \pm \sqrt{1-4}}{2}$$

$$= \frac{1 \pm \sqrt{3}i}{2}$$

$$x^3 = \frac{1 + \sqrt{3}i}{2} \quad \text{or} \quad x^3 = \frac{1 - \sqrt{3}i}{2}$$

$$= \text{cis } \pi/3 \quad \quad \quad = \text{cis } -\pi/3$$

$$x_k = \text{cis } \frac{2k\pi + \pi/3}{3}, \quad k=0,1,2 \quad x^3 = \text{cis } \frac{-2k\pi - \pi/3}{2}, \quad k=0,1,2$$

$$x_1 = \text{cis } \pi/9$$

$$x_4 = \text{cis } -\pi/9$$

$$x_2 = \text{cis } 7\pi/9$$

$$x_5 = \text{cis } -7\pi/9$$

$$x_3 = \text{cis } 13\pi/9$$

$$x_6 = \text{cis } -13\pi/9$$

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$$(ii) \quad \alpha + \beta = 2 \cos \frac{\pi}{9} \quad \alpha \beta = 1$$

$$r + s = 2 \cos \frac{7\pi}{9} \quad rs = 1$$

$$e + f = 2 \cos \frac{13\pi}{9} \quad ef = 1$$

$$\therefore x^6 - x^3 + 1 = (x^2 - 2x \cos \frac{\pi}{9} + 1)(x^2 - 2x \cos \frac{7\pi}{9} + 1)(x^2 - 2x \cos \frac{13\pi}{9} + 1)$$

(iii) Equating coeff of x^3

$$0 = -2x \cos \frac{\pi}{9} - 2x \cos \frac{7\pi}{9} - 2x \cos \frac{13\pi}{9}$$

$$\therefore \cos \frac{\pi}{9} + \cos \frac{7\pi}{9} + \cos \frac{13\pi}{9} = 0$$

Equating coeff of x^3

$$-1 = -2 \cos \frac{\pi}{9} - 2 \cos \frac{7\pi}{9} - 2 \cos \frac{13\pi}{9} - 8 \cos \frac{\pi}{9} \cos \frac{7\pi}{9} \cos \frac{13\pi}{9}$$

$$-1 = 0 - 8 \cos \frac{\pi}{9} \cos \frac{7\pi}{9} \cos \frac{13\pi}{9}$$

$$\therefore \cos \frac{\pi}{9} \cos \frac{7\pi}{9} \cos \frac{13\pi}{9} = \frac{1}{8}$$